The answer is simple. The origin of time may be similar to that of heat: it comes from averages of many microscopic variables. Let's look at this in detail.

That there is a link between time and temperature is an ancient and recurrent idea. If you think about it, all phenomena where we detect the passage of time are co-involved with temperature. The salient characteristic of time is that it moves forwards and not backwards, that is to say, there are *irreversible phenomena*. Mechanical phenomena – ones that don't involve heat – are reversible. If we film them and then run the film backwards, we see something realistic. If we film a swinging pendulum, or a stone thrown upwards then falling, and then watch the film in reverse, we still see a plausible pendulum swinging, or a stone rising and dropping to the ground.

When the stone reaches the ground, it stops, you might object: if you watch the film reversed, you see a stone leaping up from the ground by itself, and this is implausible. But when the stone reaches the ground and stops, where does its energy go? It *heats* the ground! At the precise moment when heat is produced, the process is irreversible: the past differs from the future. It is always heat and only heat that distinguishes the past from the future.

This is universal. A burning candle is transformed into smoke – the smoke cannot transform into a candle – and a candle produces heat. A boiling-hot cup of tea cools down and does not heat up: it diffuses heat. We live and get old: producing heat. Our old bicycle wears out with time: producing heat through friction. Think of the solar system. At first approximation, it continues to turn like an immense mechanism always equal to itself. It doesn't produce heat and, in fact, if you watched it in reverse you wouldn't notice anything strange about it. But looked at more closely, there are also irreversible phenomena: the Sun is using up its combustible hydrogen and will eventually exhaust it and extinguish: the Sun, too, is getting older and, in fact, produces heat. The Moon also appears to orbit the Earth unchangingly and be always equal to itself, whereas in reality it is slowly moving away. This is because it raises tides, and the tides heat the sea a little, thus exchanging energy with the Moon. Whenever you consider a phenomenon certifying the passage of time, it is through the production of heat that it does so. There is no preferred direction of time without heat.

But heat is our way to name averages over many variables.

The idea of thermal time reverses this observation. That is to say, instead of enquiring how time produces dissipation in heat, it asks how heat produces time.

Thanks to Boltzmann, we know that the notion of heat comes from the fact that we interact with averages. The idea of thermal time is that the notion of time, too, comes from the fact that we interact only with averages of many variables.[fn53]